HOW TO CONNECT SPARK TO YOUR OWN DATASOURCE

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MongoDB

SPARK SUMMIT
EUROPE 2016

mongoDB
FOR GIANT IDEAS

Spark

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MongoDB Spark connector timeline

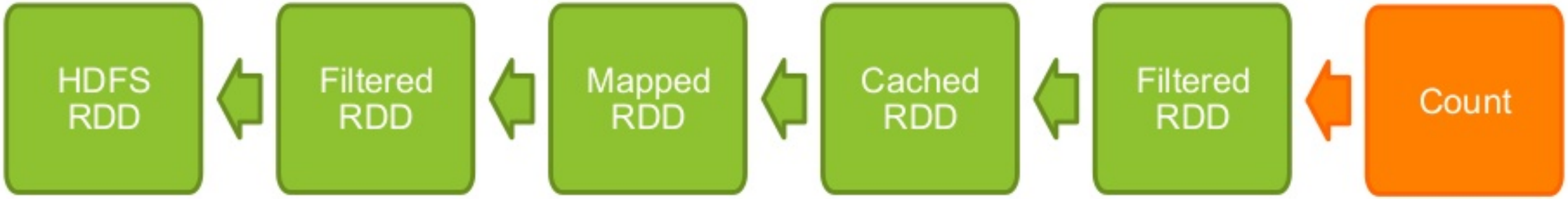
- Initial interest
  - Skunkworks project in March 2015
  - Introduction to Big Data with Apache Spark
  - Intern project in Summer 2015

- Official Project started in Jan 2016
  - Written in Scala with very little Java needed
  - Python and R support via SparkSQL
  - Much interest internally and externally

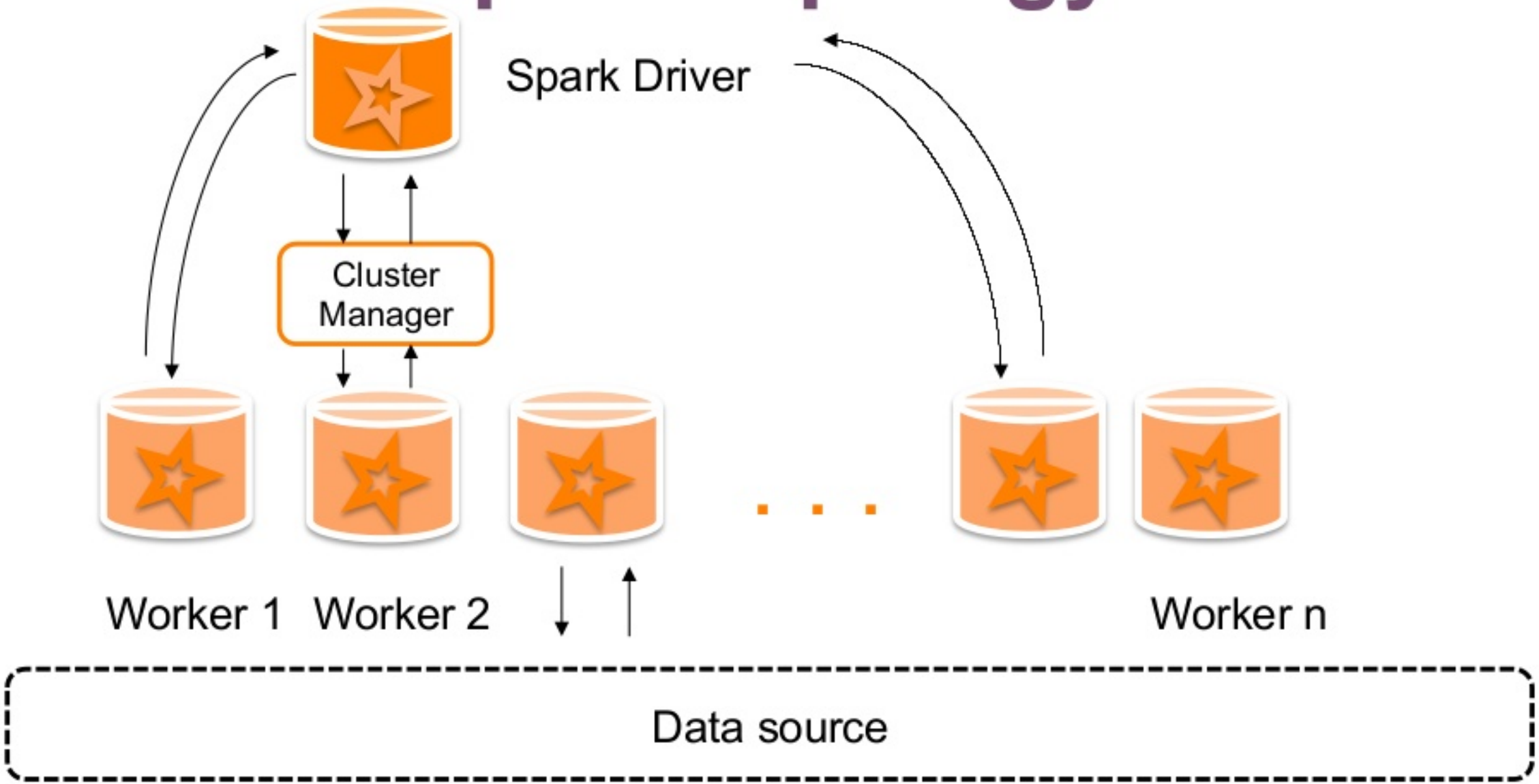
Spark 101 - the core

RDDs maintain *lineage* information that can be used to reconstruct lost partitions

```
val searches = spark.textFile("hdfs://...")
                    .filter(_.contains("Search"))
                    .map(_.split("\t")(2)).cache()
                    .filter(_.contains("MongoDB"))
                    .count()
```



Spark topology



Spark & mongoDB

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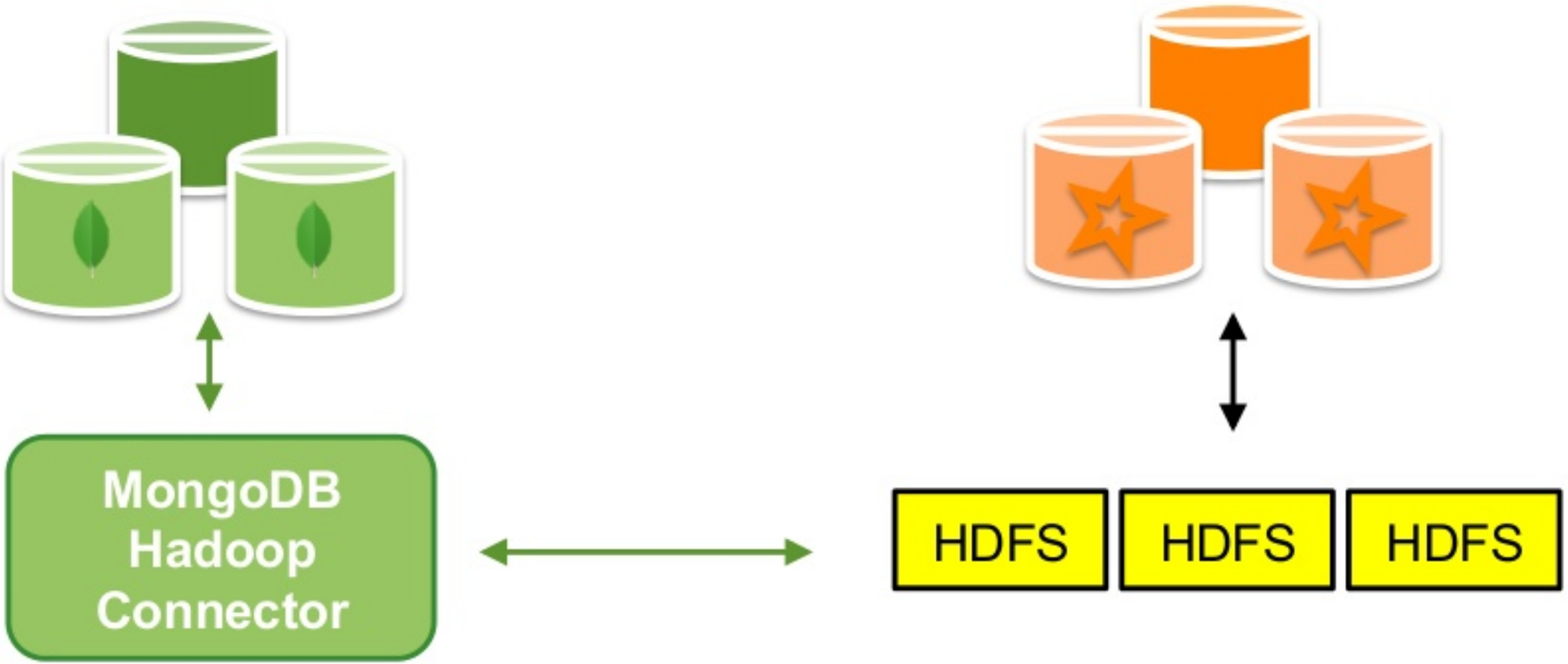
Prior to the Spark Connector



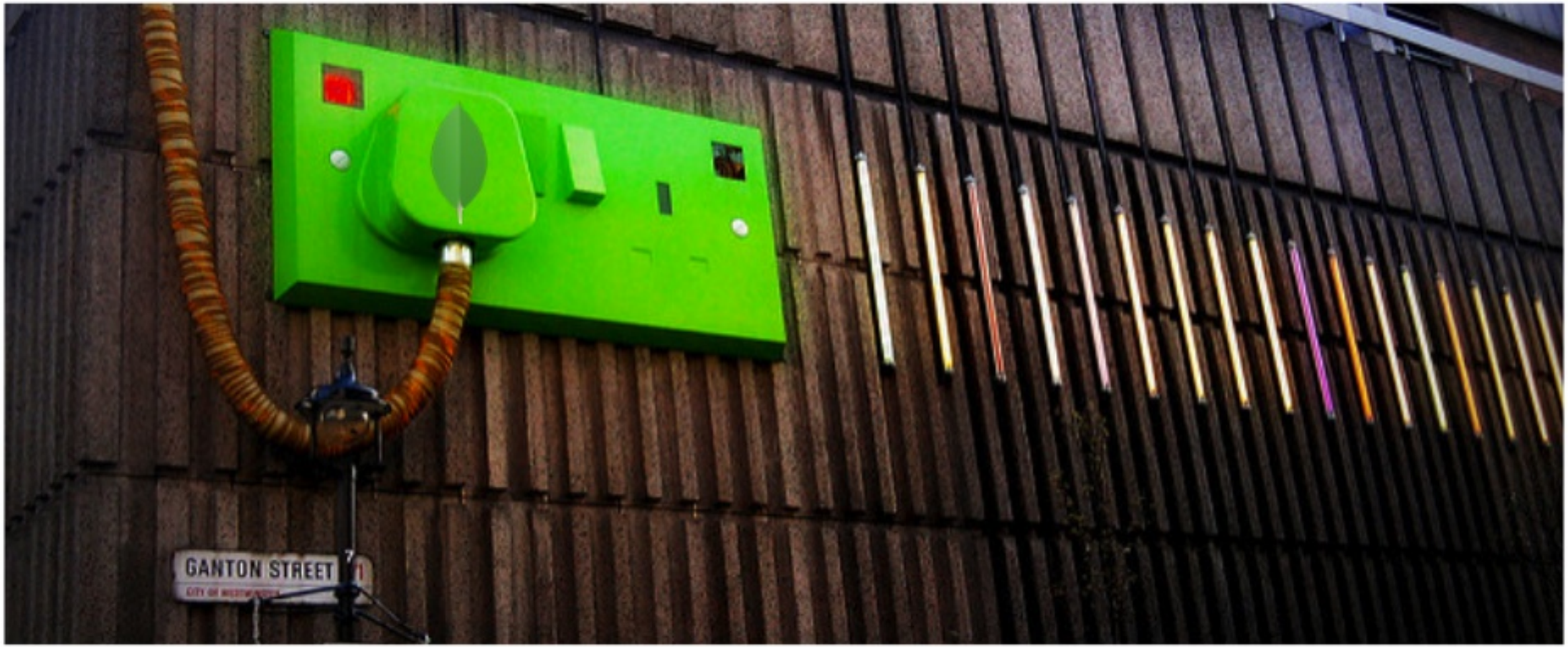
The MongoDB Spark Connector

Roll your own connector

The Golden rule

Learn from other connectors

1. Connecting to your data source



https://www.flickr.com/photos/memake/5506361372

Making a connection

- Has a cost
  - The Mongo Java Driver runs a connection pool
    Authenticates connections, replica set discovery etc..
- There are two modes to support
  - Reading
  - Writing

Connections need configuration

- **Read Configuration**
  - URI, database name and collection name
  - Partitioner
  - Sample Size (for inferring the schema)
  - Read Preference, Read Concern
  - Local threshold (for choosing the MongoS)

- **Write Configuration**
  - URI, database name and collection name
  - Write concern
  - Local threshold (for choosing the MongoS)

Connector

```
case class MongoConnector(mongoClientFactory: MongoClientFactory) extends
    Logging with Serializable with Closeable {

  def withMongoClientDo[T](code: MongoClient => T): T = {
    val client = acquireClient()
    try {
      code(client)
    } finally {
      releaseClient(client)
    }
  }


  def withDatabaseDo[T](config: MongoCollectionConfig, code: MongoDatabase => T): T =
    withMongoClientDo({ client => code(client.getDatabase(config.databaseName)) })

  def withCollectionDo[D, T](config: MongoCollectionConfig, code: MongoCollection[D] => T) …

  private[spark] def acquireClient(): MongoClient = mongoClientCache.acquire(mongoClientFactory)
  private[spark] def releaseClient(client: MongoClient): Unit = mongoClientCache.release(client)
}
```

Connection Optimization

- MongoClientCache
  - A lockless cache for MongoClients.
  - Allowing multiple tasks access to a MongoClient.
  - Keeps clients alive for a period of time.
  - Timeouts and closes old.

Making a connection

- Should be cheap as possible
  - Broadcast it so it can be reused.
  - Use a timed cache to promote reuse and ensure closure of resources.
- Configuration should be flexible
  - Spark Configuration
  - Options – Map[String, String]
  - ReadConfig / WriteConfig instances

2. Read data

https://www.flickr.com/photos/ericejohnson/4408073661/

Implement RDD[T] class

- Partition the collection
- Optionally provide preferred locations of a partition
- Compute the partition into an Iterator[T]

Partition

```scala
/**
 * An identifier for a partition in an RDD.
 */
trait Partition extends Serializable {
  /**
   * Get the partition's index within its parent RDD
   */
  def index: Int

  // A better default implementation of HashCode
  override def hashCode(): Int = index

  override def equals(other: Any): Boolean = super.equals(other)
}
```

MongoPartition

Very simple, stores the information on how to get the data.

```
case class MongoPartition(
    index: Int,
    queryBounds: BsonDocument,
    locations: Seq[String]) extends Partition
```

MongoPartitioner

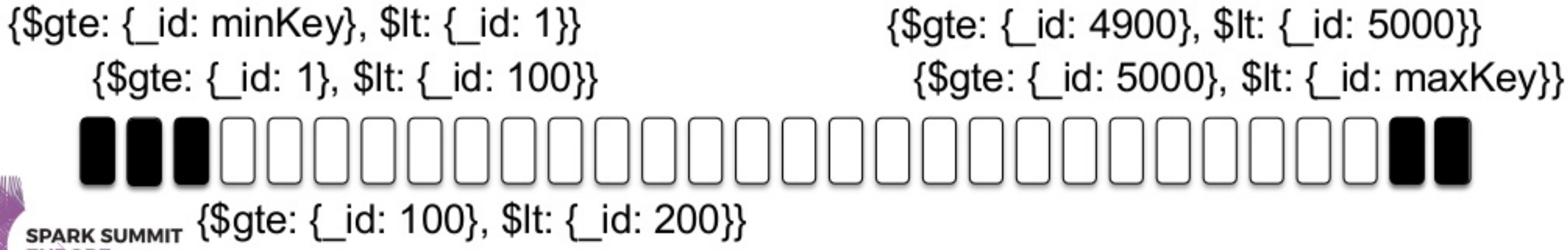
```scala
/**
 * The MongoPartitioner provides the partitions of a collection
 */
trait MongoPartitioner extends Logging with Serializable {

  /**
   * Calculate the Partitions
   *
   * @param connector the MongoConnector
   * @param readConfig the [[com.mongodb.spark.config.ReadConfig]]
   * @return the partitions
   */
  def partitions(connector: MongoConnector,
                 readConfig: ReadConfig,
                 pipeline: Array[BsonDocument]): Array[MongoPartition]

}
```

MongoSamplePartitioner

- Over samples the collection
  - Calculate the number of partitions.
    Uses the average document size and the configured partition size.
  - Samples the collection, sampling *n* number of documents per partition
  - Sorts the data by partition key
  - Takes each *n* partition
  - Adds a min and max key partition split at the start and end of the collection



MongoShardedPartitioner

- Examines the shard config database
  - Creates partitions based on the shard chunk min and max ranges
  - Stores the Shard location data for the chunk, to help promote locality
  - Adds a min and max key partition split at the start and end of the collection

{$gte: {_id: minKey}, $lt: {_id: 1}}

{$gte: {_id: 1000}, $lt: {_id: maxKey}}

Alternative Partitioners

- **MongoSplitVectorPartitioner**
  A partitioner for standalone or replicaSets. Command requires special privileges.

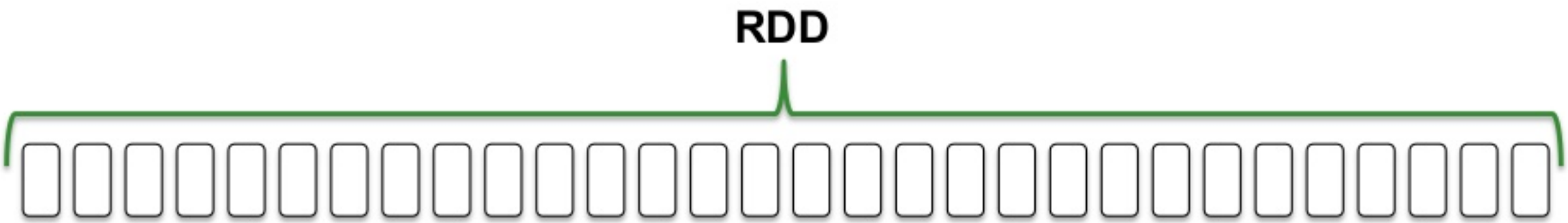
- **MongoPaginateByCountPartitioner**
  Creates a maximum number of partitions
  Costs a query to calculate each partition

- **MongoPaginateBySizePartitioner**
  As above but using average document size to determine the partitions.

- **Create your own**
  Just implement the MongoPartitioner trait and add the full path to the config

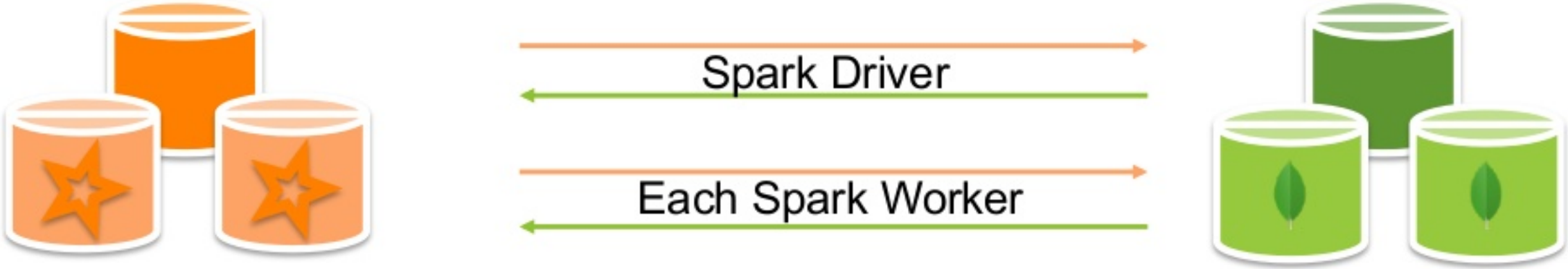
Partitions

- They are the foundations for RDD's
- Super simple concept
- Challenges for mutable data sources as not a snapshot in time



Reads under the hood

```
MongoSpark.load(sparkSession).count()
```



1. Create a MongoRDD[Document]
2. Partition the data
3. Calculate the Partitions
4. Get the preferred locations and allocate workers
5. For each partition:
   i. Queries and returns the cursor
   ii. Iterates the cursor and sums up the data
6. Finally, the Spark application returns the sum of the sums.

Reads

- Data **must** be serializable
- Partitions provide parallelism
- Partitioners should be configurable

  No one size fits all
- Partitioning strategy may be non obvious
  - Allow users to solve partitioning in their own way

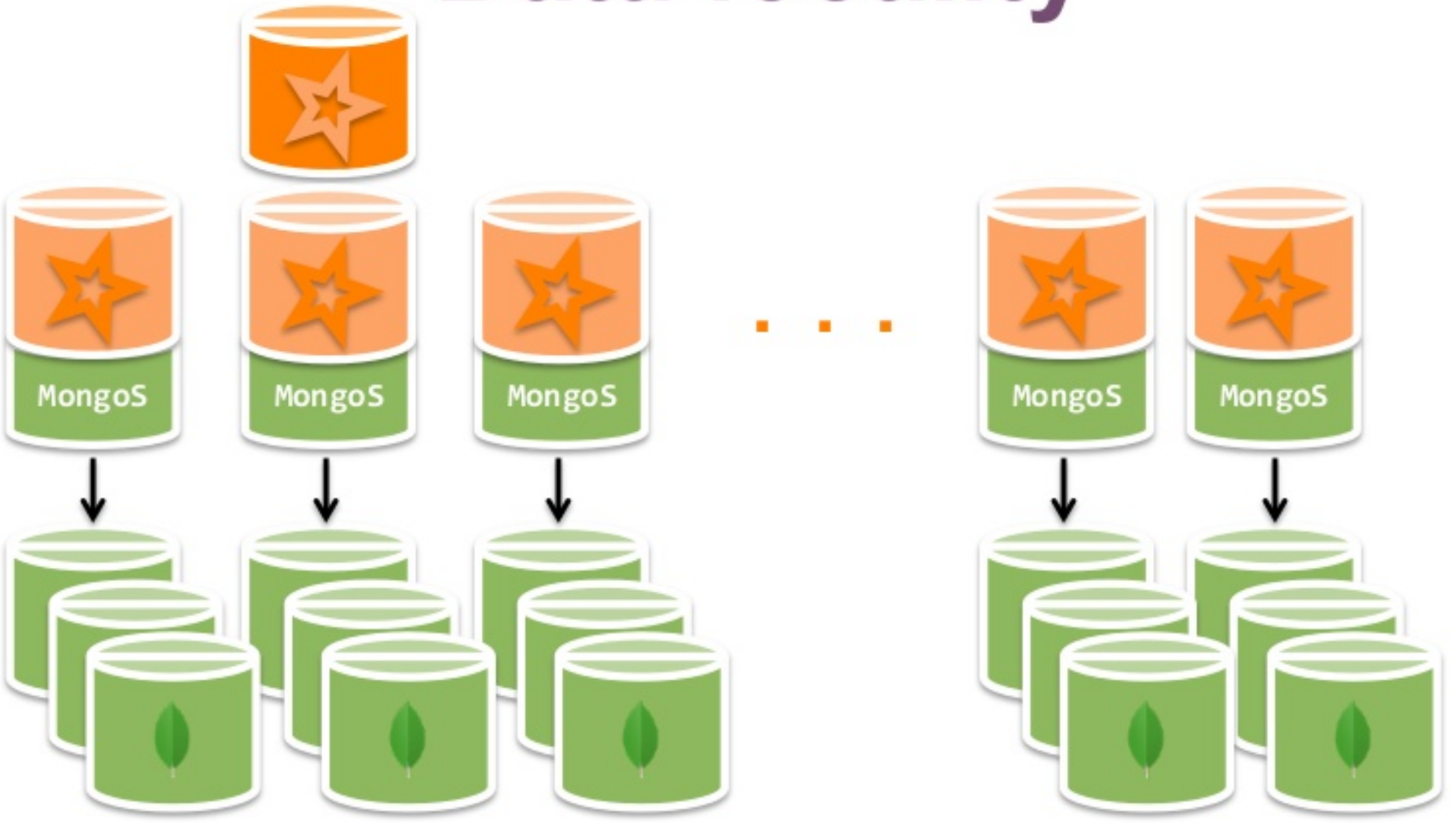
Read Performance

- MongoDB Usual Suspects
  - Document design
  - Indexes
  - Read Concern
- Spark Specifics
  - Partitioning Strategy
  - Data Locality

Data locality

Data locality

Data locality



Configure: LocalThreshold, MongoShardedPartitioner

Data locality

Configure: ReadPreference, LocalThreshold, MongoShardedPartitioner

3. Writing data

https://www.flickr.com/photos/knightjh/3415926549

Writes under the hood

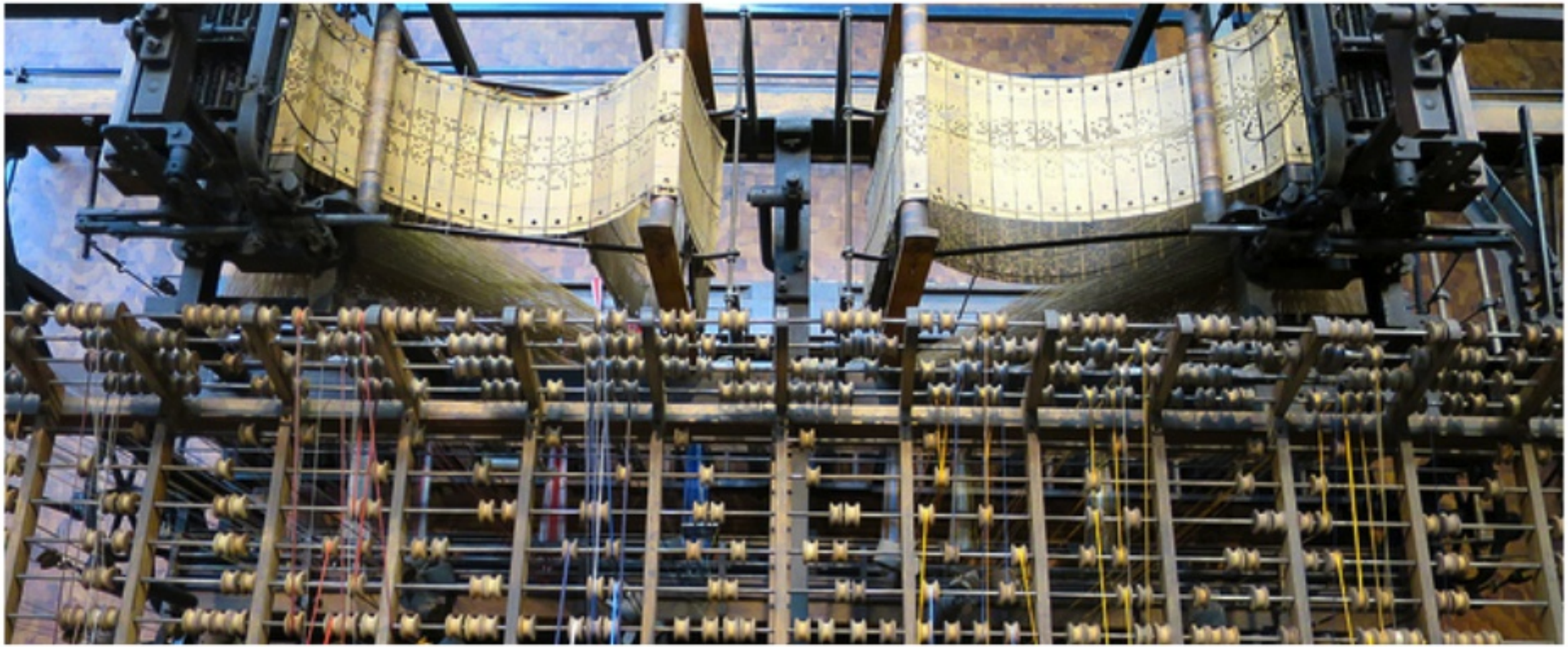
```
MongoSpark.save(dataFrame)
```

1. Create a connector
2. For each partition:
   1. Group the data in batches
   2. Insert into the collection
      * DataFrames / Datasets will upsert if there is an `_id`

LEVEL 1 COMPLETE!

GET READY FOR THE NEXT LEVEL

4. Structured data



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Why support structured data?

- RDD's are the core to Spark
  You can convert RDDs to Datasets – but the API can be painful for users
- Fastest growing area of Spark
  40% of users use DataFrames in production & 67% in prototypes*
- You can provide Python and R support
  62% of users use Spark with Python, behind Scala but gaining fast*
- Performance improvements
  Can passing filters and projections down to the data layer

* Figures from the Spark 2016 survey

Structured data in Spark

- DataFrame == Dataset[Row]
  - RDD[Row]
    - Represents a row of data
    - Optionally define the schema of the data
- Dataset
  - Efficiently decode and encode data

Create a DefaultSource

- DataSourceRegister
  Provide a shortname for the datasource (broken? not sure how its used)
- RelationProvider
  Produce relations for your data source – inferring the schema (read)
- SchemaRelationProvider
  Produce relations for your data source using the provided schema (read)
- CreatableRelationProvider
  Creates a relation based on the contents of the given DataFrame (write)

DefaultSource continued…

- StreamSourceProvider (experimental)
  Produce a streaming source of data
  Not implemented - In theory could tail the OpLog or a capped collection as a source of data.
- StreamSinkProvider (experimental)
  Produce a streaming sink for data

Inferring Schema

- Provide it via a Case Class or Java bean
  Uses Sparks reflection to provide the Schema
- Alternatively Sample the collection
  - Sample 1000 documents by default
  - Convert each document to a StructType
    For unsupported Bson Types we use extended json format
    Would like support for User Defined Types
  - Use treeAggregate to find the compatible types
    Uses TypeCoercion.*findTightestCommonTypeOfTwo* to coerce types
    Conflicting types log a warning and downgrade to StringType

Create a BaseRelation

- TableScan
  Return all the data
- PrunedScan
  Return all the data but only the selected columns
- PrunedFilteredScan
  Returns filtered data and the selected columns
- CatalystScan
  Experimental access to logical execution plan
- InsertableRelation
  Allows data to be inserted via the INSERT INTO

Multi-language support!

```
// Scala
sparkSession.read.format("com.mongodb.spark.sql").load()

// Python
sqlContext.read.format("com.mongodb.spark.sql").load()

// R
read.df(sqlContext, source = "com.mongodb.spark.sql")
```

Rolling your own Connector

- Review existing connectors https://github.com/mongodb/mongo-spark
- Partitioning maybe tricky but is core for parallelism.
- `DefaultSource` opens the door to other languages
- Schema inference is painful for Document databases and users
- Structured data is the future for Speedy Spark

Free, online training

http://university.mongodb.com

THANK YOU.

Any questions come see us at the booth!

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